Health VAASS CuraMind

Sector: Healthcare Analytics

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GitHub Repository : https://github.com/AmanCrafts/Health_VAASS_CuraMind

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EXECUTIVE SUMMARY

The healthcare industry generates large volumes of patient data, yet extracting actionable insights from this data remains a significant challenge. This project, Health VAASS CuraMind, focuses on analysing patient health records to uncover patterns that can improve patient outcomes, optimise treatment strategies, and support data-driven healthcare decisions.

The dataset used for this project was sourced from Kaggle and consists of raw patient health records containing demographic details, medical conditions, treatments, and outcomes. The dataset required extensive preprocessing due to missing values, inconsistent formats, and categorical variations. A complete ETL pipeline was implemented in Python using Jupyter Notebooks, ensuring data cleaning, transformation, and reproducibility.

Exploratory Data Analysis (EDA) was conducted to identify trends, distributions, and relationships across variables. Statistical techniques such as correlation analysis and regression were applied to uncover deeper insights into the factors influencing patient health outcomes.

An interactive Tableau dashboard was developed to allow stakeholders to explore key metrics dynamically. The dashboard provides a decision-support system by highlighting patterns across demographics, treatment effectiveness, and condition distribution.

Key insights revealed that certain demographic groups are more prone to specific conditions, while treatment effectiveness varies significantly across patient segments. Based on these findings, actionable recommendations were proposed to improve healthcare resource allocation, preventive care strategies, and treatment planning.

This project demonstrates the ability to perform end-to-end data analytics, combining technical implementation with business-focused insights to support better decision-making in the healthcare sector.

PROBLEM STATEMENT & OBJECTIVES

Problem Statement:
The primary problem addressed in this project is the lack of actionable insights from raw healthcare data, which limits the ability of healthcare providers to make informed decisions regarding patient care and resource allocation.

Objectives:
- To analyse patient health data and identify key factors affecting patient outcomes
- To uncover patterns across demographic groups and medical conditions
- To evaluate treatment effectiveness across different segments
- To build a data-driven dashboard for decision support

Scope:
The project focuses on analysing structured patient health records and deriving insights using statistical and visualisation techniques. Advanced predictive modelling is considered out of scope.

Success Criteria:
- Clean and analysis-ready dataset
- Meaningful insights derived from EDA and statistical analysis
- Functional Tableau dashboard with decision-making capabilities
- Actionable business recommendations

DATASET DESCRIPTION

The dataset used in this project was sourced from kaggle. It contains approximately 10,000 rows and 17 columns.
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ETL & DATA CLEANING

The dataset used in this project was sourced from Kaggle:
https://www.kaggle.com/datasets/mursaleen880/patient-health-record-raw-csv

It consists of over 5000+ records and multiple columns capturing patient demographics, medical conditions, and treatment-related information.

Key columns include:
- Age
- Gender
- Medical Condition
- Treatment Type
- Admission Details
- Outcome Indicators

The dataset represents raw healthcare records and contains several data quality issues such as missing values, inconsistent formats, and duplicate entries.

Limitations:
- Dataset may not represent all geographical regions
- Some fields contain missing or incomplete information
- Limited variables for advanced predictive modelling

Despite these limitations, the dataset is suitable for exploratory and statistical analysis.

KPI & METRICS

Key Performance Indicators (KPIs) were defined to measure healthcare performance and support decision-making.

KPIs include:
- Patient Recovery Rate: Measures the percentage of patients who recovered successfully
- Average Treatment Duration: Indicates efficiency of healthcare processes
- Condition Distribution: Shows prevalence of diseases across population
- Demographic Analysis: Highlights trends across age and gender

These KPIs are directly aligned with the project objectives and provide measurable indicators of healthcare system performance.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was performed to identify patterns, trends, and anomalies in the dataset.

Key analyses included:
- Distribution of patients across age groups
- Frequency of medical conditions
- Comparison of treatment outcomes across demographics
- Identification of outliers in key variables

Insights:
- Certain age groups show higher susceptibility to specific conditions
- Some treatments demonstrate higher effectiveness rates
- Data reveals variation in outcomes across different demographic segments

Visualisations such as bar charts, histograms, and box plots were used to support analysis.

STATISTICAL ANALYSIS

Statistical analysis was conducted to uncover deeper relationships within the data.

Methods used:
- Correlation analysis to identify relationships between variables
- Regression analysis to understand predictive relationships
- Hypothesis testing to validate significant differences

Findings:
- Strong relationships were observed between age and condition severity
- Certain treatments showed statistically significant differences in outcomes
- Demographic factors influence recovery rates

These results provide a deeper understanding of healthcare patterns beyond basic analysis.

KEY INSIGHTS

1. Patients in specific age groups are more vulnerable to certain medical conditions.
2. Treatment effectiveness varies significantly across different patient segments.
3. Certain conditions show higher prevalence in specific demographics.
4. Recovery rates are influenced by both demographic and treatment factors.
5. Some treatments consistently outperform others in terms of success rates.
6. Data highlights inefficiencies in treatment duration for certain conditions.
7. High-risk groups can be identified using demographic indicators.
8. Preventive healthcare strategies can significantly reduce case severity.

BUSINESS RECOMMENDATIONS

1. Focus healthcare resources on high-risk demographic groups to improve outcomes.
2. Optimise treatment strategies based on effectiveness analysis.
3. Implement preventive care programs targeting vulnerable populations.
4. Improve data collection processes for better analytics in future.
5. Use data-driven insights to enhance decision-making in healthcare management.

LIMITATIONS & FUTURE SCOPE

Limitations:
- Dataset limitations may affect generalisation
- Missing values may impact accuracy
- Limited scope for predictive modelling

Future Scope:
- Apply machine learning models for prediction
- Use real-time healthcare data
- Expand dataset for broader analysis

CONTRIBUTIONS

Shubhi Kumari: Cleaning, Preprocessing, EDA, Statistical Analysis
Amanjeet Malik: Cleaning, Preprocessing, EDA, Statistical Analysis
Vansh Dagar: Tableau Dashboard
Anishka Khurana: Tableau Dashboard
Shorya Taneja: Report Writing